📘 Excel Data Cleaning & Validation Documentation

Module: Data Cleaning, Validation & Error Handling in Excel

Using **Microsoft Excel**

🎯 Learning Objectives

By the end of this session, learners will be able to:

- Identify and remove duplicate data
- Split and organize text using Text to Columns
- Use Flash Fill for automation
- Create Data Validation rules
- Detect and correct errors in datasets
- Improve data quality for analysis

🔹 Introduction to Data Cleaning

In real-world datasets:
❌ Duplicate records
❌ Wrong formats
❌ Missing values
❌ Typing mistakes
❌ Invalid entries

are very common.

🔥 Why Data Cleaning is Important?

Bad data leads to:

- Wrong reports
- Incorrect dashboards
- Poor business decisions

A. Identifying and Removing Duplicates

1️⃣ What are Duplicates?

Duplicate data means:
👉 Same record appears multiple times.

🔹 Example

Emp_ID	Name
101	Ravi
102	Anita
101	Ravi

Here:

```None
101 Ravi
```

appears twice.

🔥 Problems Caused by Duplicates

Duplicates can:
❌ Increase totals incorrectly
❌ Affect reports
❌ Create billing mistakes
❌ Disturb dashboards

2️⃣ How to Identify Duplicates

🔹 Method 1: Conditional Formatting

Steps:

```None
Home → Conditional Formatting → Highlight Cells Rules → Duplicate Values
```

🔹 Result

Excel highlights duplicate entries automatically.

🧪 Practice Activity

Highlight duplicate:

- Employee IDs
- Phone numbers
- Invoice IDs

3️⃣ Remove Duplicates

🔹 Steps

1. Select dataset
2. Go to:

```
None
Data → Remove Duplicates
```

3. Select columns
4. Click:

```
None
OK
```

🔥 Example

Before:

Emp_ID
101
102
101

After:

Emp_ID
101
102

🔥 Real-Time Business Example

In hospitals:
Duplicate patient IDs may:
❌ Duplicate billing
❌ Duplicate appointments

Removing duplicates ensures accurate reporting.

B. Text to Columns and Flash Fill

1️⃣ Text to Columns

Used to split combined text into multiple columns.

🔹 Example

Full Name
Tota Jagadeesh

Split into:

First Name	Last Name

Tota	Jagadeesh

🔹 Steps

1. Select column
2. Go to:

```
Data → Text to Columns
```

3. Choose:

```
Delimited
```

4. Select:

```
Space
```

5. Finish

🔥 Result

Excel separates names automatically.

🔥 Real-Time Use Cases

Used for:

- Full names

- Addresses
- Email IDs
- CSV files

2️⃣ Flash Fill

Flash Fill automatically detects patterns.

🔹 Example

Full Name	First Name
Ravi Kumar	Ravi

Type manually:

```None
Ravi
```

Then press:

```None
Ctrl + E
```

Excel auto-fills remaining names.

🔥 Flash Fill Examples

Extract First Name

From:

```None
Ravi Kumar
```

to:

```None
Ravi
```

Extract Email Username

From:

```None
ravi@gmail.com
```

to:

```None
ravi
```

🔥 Advantages of Flash Fill

✅ Fast
✅ No formulas needed
✅ Smart pattern detection

🧪 Practice Activity

Use Flash Fill to:

- Extract first names
- Create email IDs

- Convert text formats

C. Data Validation Rules

🔹 What is Data Validation?

Data Validation controls:
👉 What users can enter into cells.

🔥 Why Important?

Prevents:
❌ Wrong entries
❌ Invalid data
❌ Typing mistakes

🔹 Access Path

```None
Data → Data Validation
```

1️⃣ List Validation

Creates dropdown lists.

🔹 Example

Allow only:

- HR
- IT
- Finance

🔹 Steps

1. Select cells
2. Data Validation
3. Allow:

```None
List
```

4. Source:

```None
HR,IT,Finance
```

🔥 Result

Dropdown appears.

🔥 Real-Time Example

Departments:

- HR
- Sales

- Finance

avoids spelling mistakes.

2️⃣ Whole Number Validation

Allows only integers.

🔹 Example

Age:

```None
18 to 60
```

🔹 Steps

Allow:

```None
Whole Number
```

Condition:

```None
Between 18 and 60
```

🔥 Result

Invalid ages rejected.

3️⃣ Date Validation

Restricts date entries.

🔹 Example

Allow dates only for:

```
2026
```

🔹 Use Cases

- Joining date
- Invoice date
- Appointment date

4️⃣ Custom Validation

Uses formulas for advanced rules.

🔥 Example

Allow only values starting with:

```None
EMP
```

Formula:

```None
=LEFT(A1,3)="EMP"
```

🔥 Real-Time Example

Employee IDs:

```None
EMP101
```

🧪 Practice Activity

Create validations for:
✅ Department dropdown
✅ Age limit
✅ Date restriction
✅ Employee ID format

D. Error Checking and Correction Techniques

🔹 Common Excel Errors

Error	Meaning
#DIV/0!	Divide by zero
#VALUE!	Wrong data type
#NAME?	Wrong formula name
#REF!	Invalid reference
#N/A	Value not found

1️⃣ #DIV/0! Error

🔹 Example

```None
=10/0
```

Result:

```None
#DIV/0!
```

🔹 Solution

Use:

```None
=IF(B2=0,"",A2/B2)
```

2️⃣ #VALUE! Error

🔹 Example

Trying to calculate:

```None
Text + Number
```

🔹 Solution

Check:

- Data types
- Cell contents

3️⃣ #NAME? Error

🔹 Example

```None
=SU(A1:A5)
```

❌ Wrong function name.

Correct:

```None
=SUM(A1:A5)
```

4️⃣ #REF! Error

🔹 Meaning

Reference deleted or invalid.

🔹 Example

Deleting referenced column.

🔹 Solution

Correct cell references.

5️⃣ Error Checking Tool

🔹 Path

```None
Formulas → Error Checking
```